MIL-R-52305A(ME)
8 November 1973
SUPERSEDING
MIL-R-52305(MO)
5 June 1963

MILITARY SPECIFICATION

ROLLER ASSEMBLY, BRIDGE-LAUNCHING: 40-TON-CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a bridge-launching roller assembly designed for a maximum load of 40 tons.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

FSC 5420

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Military

MIL-P-116	-Preservation-Packaging, Methods of.
MIL-T-704	-Treatment and Painting of Materiel.
MIL-G-10924	-Grease, Automotive and Artillery.
MIL-G-23827	-Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS

Military

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-Marking for Shipment and Storage.
MIL-STD-130	-Identification Marking of US Military Property.

DRAWINGS

ME

TA13208E4590	-Roller, Bridge, Launching, 40-Ton-Capacity Assembly.

(Copies of specifications, standards, and drawings required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN WELDING SOCIETY

A5.5	-Low-Alloy Steel Covered Arc-Welding Electrodes
D1.1	-Structural Welding Code

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The roller assembly shall be in accordance with the Top Assembly TA13208E4590 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

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3.2 First article (preproduction model). The supplier shall furnish one or more roller assemblies as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production, that his production methods will produce roller assemblies that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings.

3.4 Welding electrodes. Welding electrodes shall be in accordance with American Welding Society (AWS) A5.5.

3.5 Lubrication. The support and alining roller-bearings shall be lubricated with greases conforming to MIL-G-10924 or MIL-G-23827. Each roller assembly shall be tagged in a conspicuous place to indicate the temperature range and grade of grease used.

3.6 Treatment and painting. The roller assembly, except for pins, bolts, nuts, and bearings, shall be cleaned, treated, and painted in accordance with MIL-T-704, type B. Pins, bolts, and nuts shall be coated as specified in 5.2.1.

3.7 Identification marking. The roller shall be identified in accordance with MIL-STD-130. Marking shall be applied by metal stamping.

3.8 Workmanship.

3.8.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken

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to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.8.2 Bolt holes. Bolt holes shall be accurately formed and shall have the burrs removed.

3.8.3 Welders and welding.

3.8.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by that code:

Structural Welding Code of the American Welding Society, AWS D1.1

3.8.3.2 Welding. The surface of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings. Welding shall be done by the shielded metal-arc method in accordance with the applicable requirements of the American Welding Society "Structural Welding Code". All welded parts shall be free from cracks and other imperfections of the part. Work shall be positioned for flat welding, whenever practicable.

3.8.3.2.1 Jigs and fixtures. Components shall be assembled in steel jigs or fixtures and joined while held in position. Jigs or fixtures shall be designed to prevent distortion or to limit the distortion to an amount that can be removed by straightening and stress relieving without changing the physical characteristics of the metal.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The roller assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each roller assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The roller assemblies shall be examined as specified herein, for the following defects:

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101. Materials not as specified.
102. Nonconformance to dimensions shown on the drawings.
103. Incomplete assembly.
104. Weld defects, such as base-metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, slag inclusions, or spatter.
105. Rollers not cylindrical or not free to rotate.
106. Components containing projections or burrs.
107. Treatment and painting not as specified.
108. Lubrication and lubrication fittings not as specified.
109. Identification markings incorrect, illegible, or missing.
110. Workmanship not as specified.
111. Qualification of welders or weld procedure not as specified.

4.6 Inspection of preparation for delivery.

4.6.1 Preproduction-pack inspection.

4.6.1.1 Examination. Examine the preproduction pack for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

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112. Unpainted exterior metal surfaces not coated with preservative as specified for level A.
113. Boxes not as specified for level A, B, or C.
114. Strapping not zinc coated for level A.
115. Marking missing, illegible, incorrect, or incomplete for level A, B, or C.

5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. The supplier shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove prior to starting production packing, that the applied preservation, packing, and marking comply with the preparation for delivery requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be accomplished utilizing either the preproduction roller or a production roller. If the preproduction model roller is utilized, any preservation and packing shall be removed by the supplier at no expense to the Gevernment when requested by the Government to facilitate comparison between the preproduction roller and the production roller.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A. Unpainted exterior metal surfaces of the roller assembly requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 perservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.2.2 Level C. Each roller assembly shall be preserved to afford protection against deterioration and damage from the supply source to the initial destination.

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5.3 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.3.1 Level A. Each roller assembly shall be packed in a close fitting box conforming to PPP-B-601, overseas type, style A or B, grade B or PPP-B-621, class 2, style 2. The roller assembly shall be blocked, braced, and anchored within the box in a manner to prevent movement or damage. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated.

5.3.2 Level B. Each roller assembly shall be packed as specified in 5.3.1, except boxes shall be domestic type or class as applicable and strapping shall not be required to be zinc coated.

5.3.3 Level C. Each roller assembly shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking. Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The roller is intended to be used in launching beam and girder line-of-communication bridges.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of preproduction roller assemblies required and time frame required for submission of preproduction model(s) (see 3.2).
(c) Time frame required for submission of the preproduction pack (see 5.1).
(d) Level of preservation and level of packing required (see 5.2 and 5.3).

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6.3 Preproduction model. Any changes or deviations of production roller assemblies from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish roller assemblies conforming to this specification.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, pack, and mark the roller assemblies in accordance with this specification.

Custodian:

Army - ME

Preparing activity:

Army - ME

Project No. 5420-A179

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-R-52305A(ME) Roller Assembly, Bridge-Launching: 40-Ton-Capacity

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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